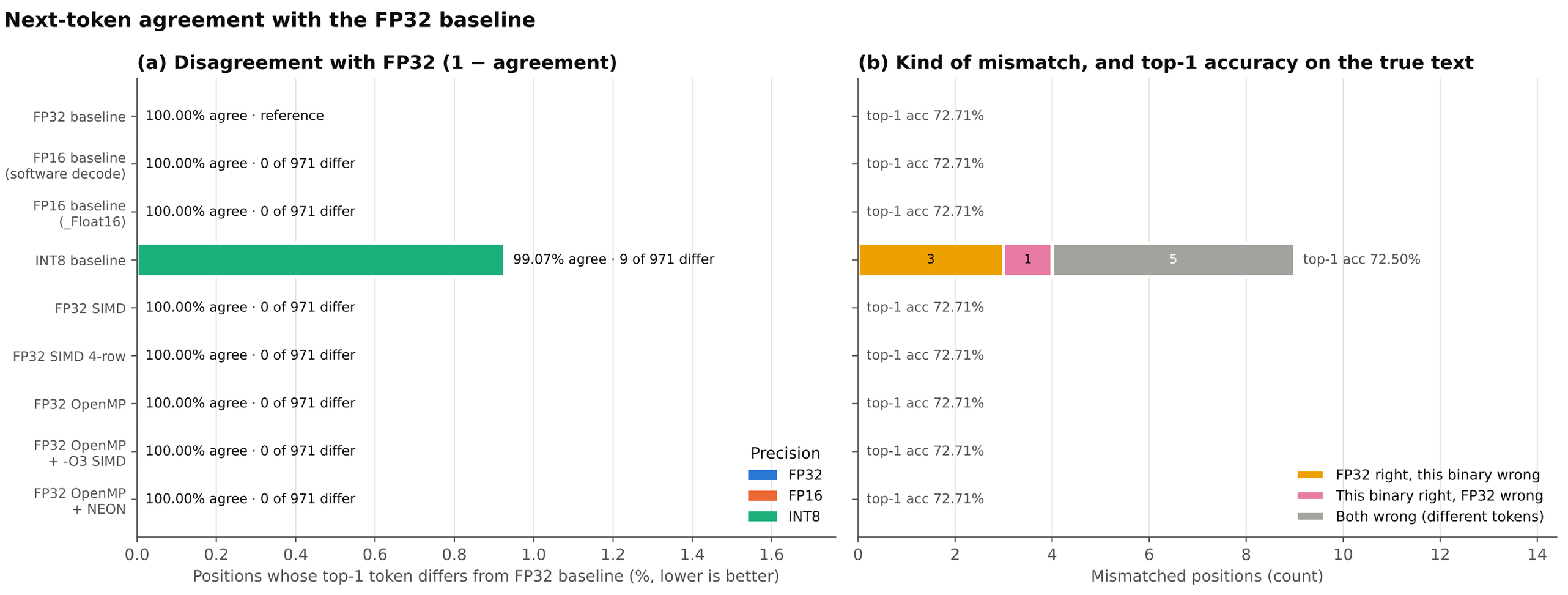
Next-token agreement with the FP32 baseline
(a) Disagreement with FP32 (1 − agreement)
FP32 baseline
100.00% agree · reference
FP16 baseline (software decode)
100.00% agree · 0 of 971 differ
FP16 baseline (_Float16)
100.00% agree · 0 of 971 differ
INT8 baseline
99.07% agree · 9 of 971 differ
FP32 SIMD
100.00% agree · 0 of 971 differ
FP32 SIMD 4-row
100.00% agree · 0 of 971 differ
FP32 OpenMP
100.00% agree · 0 of 971 differ
FP32 OpenMP + -O3 SIMD
100.00% agree · 0 of 971 differ
FP32 OpenMP + NEON
100.00% agree · 0 of 971 differ
Precision
FP32
FP16
INT8
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
Positions whose top-1 token differs from FP32 baseline (%, lower is better)
(b) Kind of mismatch, and top-1 accuracy on the true text
top-1 acc 72.71%
top-1 acc 72.71%
top-1 acc 72.71%
3
1
5
top-1 acc 72.50%
top-1 acc 72.71%
top-1 acc 72.71%
top-1 acc 72.71%
top-1 acc 72.71%
top-1 acc 72.71%
FP32 right, this binary wrong
This binary right, FP32 wrong
Both wrong (different tokens)
0
2
4
6
8
10
12
14
Mismatched positions (count)
Teacher forcing on data/eval.txt (971 positions): at each position both models see the same true context; argmax of the logits compared (temperature-independent). Top-1 accuracy = argmax equals the true next token.